

Coursework Report Template for Module INM433 “Visual Analytics”

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Abstract— In this paper we will look to analyse suicide data which contains information about suicide for different countries around the world over a period of 30 years. Temporal and spatial analysis was used to find trends and patterns. We have found Suicide rates have been decreasing since the turn of the century however is still a cause of concern, especially amongst males and the 75+ age group who are the most affected. How rich a country is has a little correlation between suicide rate and each country has their own unique pattern with suicide.

1 PROBLEM STATEMENT

Suicide is defined as an act of causing one's own death and can be caused by factors such as mental illness, substance abuse and health problems, to name a few [7]. 'It is estimated by the World health organization that approximately 817,000 have died globally by suicide, which is the equivalent to one suicide every 40 seconds' [7].

This paper will look at visual analytic analysis to give a better understanding of the trends associated with suicide around the world. The questions which will be answered are:

1. What is the overall trend in suicide during this period?
2. What has been the impact to individuals during this period?
3. Does an economic change affect the number of suicides?
4. What are the trends in suicide rate pre 2000 and post 2000?
5. What can be predicted about future number of suicides?

2 STATE OF THE ART

The below articles look at a unique feature involving suicide and present their findings differently. Paper 1 looks at suicide during the pandemic in the US between 2018-2021, paper 2 compares suicide with GDP changes and paper 3 compares suicide to other related deaths.

Paper [1] uses spatial visualization show suicide ideation and suicide attempts of different types of people categorized by 'Age', 'Sex', 'Race' and 'First Language', during the different stages e.g. 'First Case', "Post-COVID-19 First Wave", using a colored scale to dictate the number of people. For example, the higher the number the darker the color will be of the respective state. This made it possible to visualize the significant between each state of the stages of the pandemic. In addition to visualizations to confirm that there was a significant change pairwise proportionality tests were used where $P\text{-value} \leq 0.05$ considered significant. Paper concludes that Females, youth aged 15-20, White, non-Hispanic, and English speakers had increased post-pandemic suicide-related visits. This concept be particularly useful in

finding the differences between pre 2000 and post 2000 of the countries in my dataset.

Paper [2] data in this article has been collected from the WHO mortality database website and World Bank's official website. It looks at the correlation between PPP-adjusted GDP and suicide rates, across different regions. It based the correlation of regions within the world with suicide rate. It found that when a country's GDP increases, it was not always followed by a decrease in suicide rates, which would be expected, and varies between country. Developed countries (European countries, USA, Canada, New Zealand and Australia) had an inverse relationship between GDP and suicide rate. What was peculiar about this paper was the relation between developing Latin American and Caribbean countries, which saw a positive correlation between GDP and suicide. This paper then investigated factors such as health policies and infrastructure to give reasoning to why there is positive or negative correlation. What this paper did well is it categorised countries well in order to get an area understanding. I will be taking a similar approach by categorizing the countries into their respective continents; and provide a better understanding, as it would be difficult to put all the countries on the same graph and having separate graphs may make it difficult to compare.

Paper [3] starts of by giving a geographical view of the deaths, via suicide around the world, for male, female and both. And then it investigates deaths further, by providing different causes of death along with looking at the different age groups that are affected. This paper provides in-depth analysis well as it segregated the age group and compares the segregated age group with the different features. The structure of this paper was well presented as it started off with an overview before filtering down and looking at attributes in more details, an approach which I will look to follow.

3 PROPERTIES OF THE DATA

The dataset used is from Kaggle. It contains information collected from the 'United Nations Development program' and the 'World Health Organisation'. It consists of 12 columns and 27,820 rows. It contains information about 101/195 of the countries in the world and it reflects the number of suicides via

age group and sex for respective country. The age groups are divided into 6 groups; 5-15, 15-24, 25-34, 35-54, 54-74 and 75+ and sex is 2 outcomes, male or female. As well as age, the data set also contains information about the generation which the number of people are in. For example, the age groups are also categorised to their respective generation they are in, Generation X (born between 1965 and 1980), Silent (1926 and 1945), G.I Generation (Born 1901–1924), Boomers (born between 1946 to 1964), millennials (born between 1981 and 1996) and generation Z (born between 1997 and 2012).

In addition to this it also shows the population and number of suicides of the 6 age groups for the respective country between the years of 1985 to 2016. It also contains a column on the number of suicides per 100,000 of population.

Furthermore, it also includes HDI for the respective country, HDI is defined as an index of measures key dimensions of human development. These include 'life expectancy, access to education (measured by expected years of schooling of children at school-entry age and mean years of schooling of the adult population) and standard of living (measured by Gross National Income per capita adjusted for the price level of the country'[4]. Additionally, it also has information for each of the country GDP (\$). 'GDP is defined as gross domestic product and informs us of all the finished goods and services produced within a country within a time period [5], with known population we can work out GDP per capita.

Missing Values & Outliers:

Having looked at 'Nan' values we see that there are substantial missing values in the "HDI", with approximately 75% of the data missing. Due to each country's uniqueness and limited data outliers, shown in figure 1, will not be removed.

Figure 2 shows us that there is a significant drop in the number of suicides between 2015 and 2016, hence our analysis will not be including the year 2016, due to insufficient data for that year.

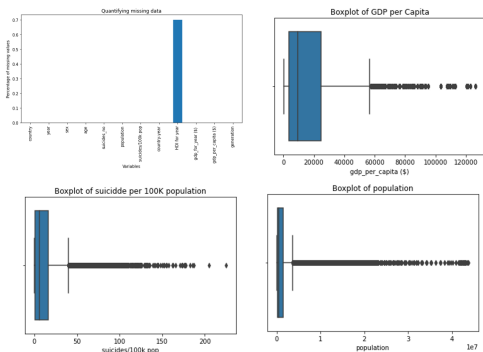
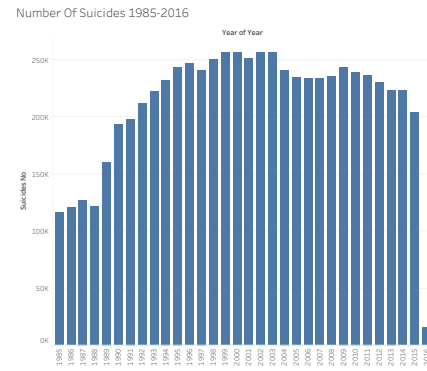


Figure 1 – Boxplots of continuous variables and bar chart of missing data.

Figure 2 – Total of suicides for each year between 1985-2016.



4 ANALYSIS

4.1 Approach

This part of the paper will involve the stages which will be carried out in order to answer the research questions, computational methods will be combined with human judgment in order to do so. Figure 3.

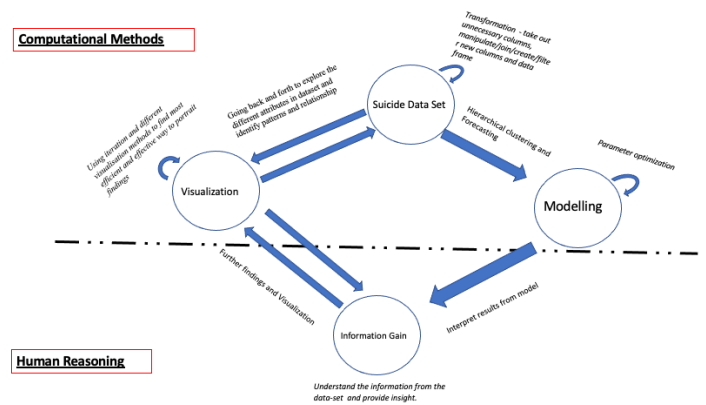


Figure 3 – Analysis workflow plan.

In order to answer question 1, we will look at Temporal analysis to look at the trend over the period and highlight any significant points such as peaks and troughs. In addition to this I will include a line to represent the average mean to see at which points of time where the suicide rate is above or below the mean. After this, I will be providing a geo-spatial map to look at the areas which are at most affected, which will involve using a colour scale to be identifying these areas. In order to understand the relationship between population and the number of suicides, Hierarchical clustering will be applied to analyse the relationship.

For question 2, I will compare the respective attribute, e.g., age group and sex and look at their respective count in this time period. Age will be used over generation as it will provide an approach which will be simpler to understand as well as provide information when people get to different stages of life. This will be shown via a bar chart representing the average

death rate and this will be accompanied by the changes over time, via time series graph.

Question 3 will involve looking at the countries which are most affected pre and post 2000, which will involve bar chart analysis, which will contain information about the most affected age group.

Question 4 will look at how the economy affects suicide rate. This will involve analysing the relationship between GDP and suicide rate, then look at points in the time period to look at correlations. In addition compare the relationship between these two factors for different regions of the world and time periods.

In order to find future predictions about suicide, from 2015 onwards, I will be using a forecasting function to do so, for each age group.

A combination of Tableau and python will be used for visualisation to answer these questions.

4.2 Process

In this part, averages per 100K population will be used as the general scale for suicide, as it will more a more in-depth analysis of the countries and make comparisons more possible. Using summed values would have given a disproportionate understanding of the statistic as well as would have given been bias to the countries with the larger population.

Looking at the overall trend, figure 4, we see that that the suicide rate per 100K has seen a slight increase from 11.5 in 1985 to 11.7 in 2016, so can infer the rate of suicide in 2015 is no different than the start of this data in 1985. From 1985 to 1995 the suicide rate increased, with sharp increases between 1988 to 1989 and 1993 to 1995. We see an overall decline from 1995 onwards with suicide rate dropping from 15.5 to 11.7, with sharp decreases from 1995 to 1997, 2004 to 2006, 2007 to 2010 and 2012 to 2015 which suggests that there must have been a change occurring in 1995 which reduced suicide. Taking the overall average into consideration, indicated by the red line in figure 4, we see that the period between 1989 and 2006 was where the suicide rate was above average, which indicated that suicide affected the population most during this period.

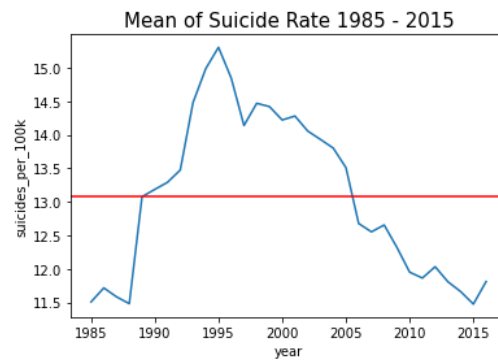


Figure 4 – Mean suicide rate for 100K population, each year

Looking at the total number of suicide deaths around the world in the time period, figure 5, we see that the Russian Federation, Kazakhstan and Lithuania are most affected by suicide in terms of its population. Upon looking into this further we see that in South America Guyana, Suriname & Uruguay are the most affected countries via suicide. In Europe, we see that Lithuania and Hungary are the most affected. Due to insufficient data, we are unable to gather an understanding on the other continents.

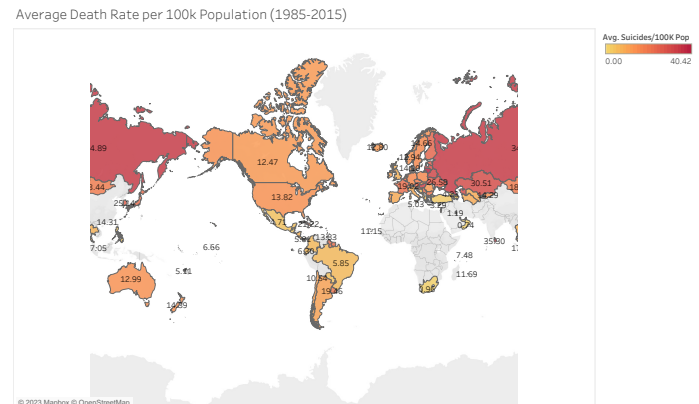


Figure 5 – Geospatial map of average suicide rate per 100k population.

Using Hierarchical clustering to look at the number of suicides and the population we see that it produces 4 distinct clusters, figure 6, with very little overlapping. Cluster 1 shows up that there is a suicide rate up to 5000 for a population of up to 0.25×10^8 , cluster 2 has a suicide rate of up to approximately 18,000 for a population between 0.25×10^8 and 0.9×10^8 . Cluster 3, which has the biggest variance in terms of suicide rate with suicide number ranging between 5,000 and 62,000 for a population between 0.9×10^8 and 1.9×10^8 . Cluster 4 had a suicide rate between 29,000 and 40,000 for a population size of 2.2×10^8 and 3×10^8 .

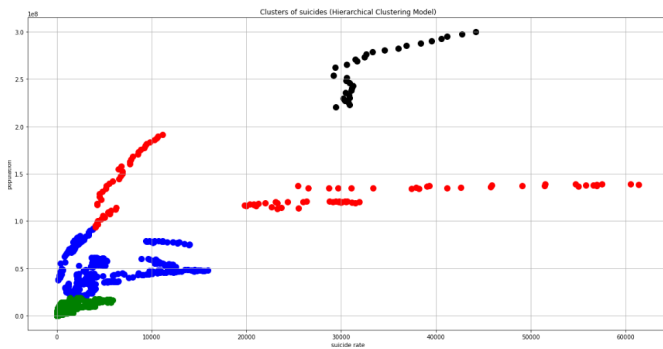


Figure 6 – Clusters of hierarchical clustering model.

We will now look at the particular type of individual who is affected. From figure 7, suicide is more amongst men than women, with it affecting men over 3 times more. Looking at the overall change between 1985 and 2015 for both sexes we see that, male suicide rate had increased from 17 to 18.5 per 100k whilst female suicide rate has showed a significant decrease from 6.3 to 5.2 per 100K of population. Due to the significant differences between the two sexes, it made sense for each sex to have their own graph so that trends will be able to identified and own comparisons will be able to be made. Furthermore, we can also see that 1995, both sexes had their peak suicide rate, which is line with figure 4, which saw 1995 as the most affected year in terms of suicide death. Additionally, looking at the bigger picture we can see that males are more volatile compared to female in this period, with males taking a range of values between 17 to 24, while women's suicide rate per 100K is between values 5.2 and 6.6 hence more compact.

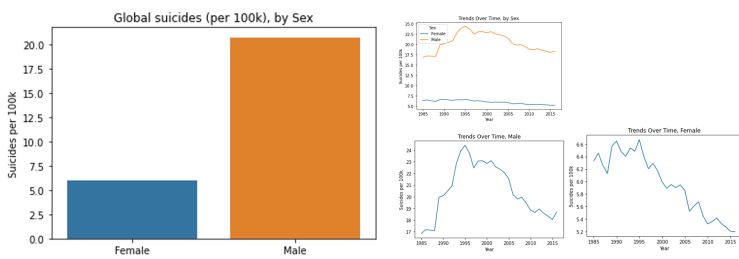


Figure 7 – On average total number of females and males affected by suicide and shows how suicide rate for both sexes changes over time.

If we look at different age groups, we see that the elderly, 75+, are the most affected age group whilst 5-14 are the least. From figure 8, we can see that as the older one gets, the suicide rate for that age group increases however there is an exponential increase in the number of deaths from 5-14 to 15-24. Looking at the trends in pattern over time we see that all age groups peak in 1995, with then showing a downwards trend after 1995. Additionally, we can see that from the start of the data, 1985, that the 75+ age group was significantly more affected compared with the other age groups, however, with time the gap decreases and becomes relatively closer to the

other age groups. Looking at the age group 5-14, we see that it relatively constant, when compared to the mean suicide rate, and is distant to the other age groups however, this could be since it is the least affected age group.

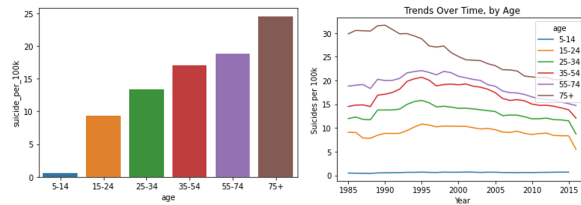
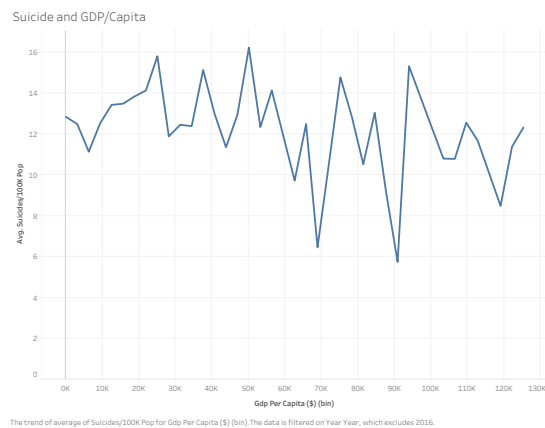


Figure 8 – Shows the average number of each age group affected per 100k population and the change over time.

In order to get a understanding between the overall relationship between GDP and suicide rate, GDP per capita was binned to help identification of trends. From figure 9, we see that there is no clear relationship between these two factors with suicide rate changing constantly when GDP changes, quite erratic. In order to analyse this further figure 10 looks at significant time points in the economy which saw radical changes in GDP, such as 'Gulf war recession in 1995', 'Dot bomb recession in 2001' and 'the great recession in 2009'[6]. From this we see that in the 1990 recession there was a slight positive correlation between GDP and suicide rate, whilst the other recession points showed no clear evidence of correlation.



The trend of average of Suicides/100K Pop for Gdp Per Capita (\$ (bin). The data is filtered on Year Year, which excludes 2016.

Figure 9- Shows the change in suicide rate as GDP changes.

Looking at countries around the world from different regions we can identify trends between GDP and suicide rate more easily, figure 10. We see that United Kingdom, United States and Uruguay have a constant suicide rate as GDP changes whereas Russia and Uruguay have contrasting effects. Russia has a negative correlation with suicide rate up to 12.5K GDP, after which suicides becomes constant. Uruguay has a positive effect with suicide rate up to 2.5k before remaining fairly constant as GDP changes.

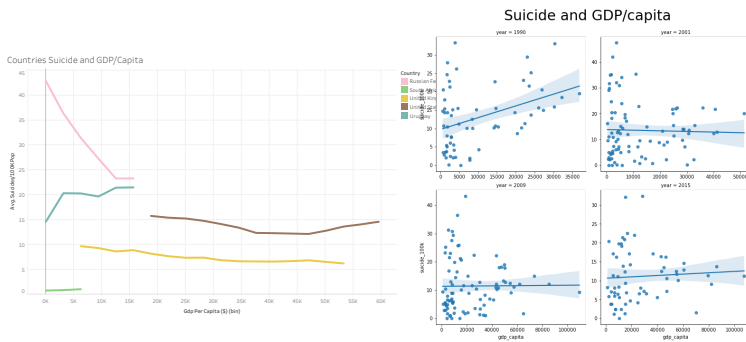


Figure 10- (Left) – Shows GDP/Suicide plot for certain countries (Right) – Shows GDP/suicide plot for points in time.

Figure 11 looks at the difference in time periods pre and post the year 2000. The data was filtered to show differences between different century's, pre 2000 (1985-1999) and post (1999-2015). Taking the average respective mean of the of the suicide rate for the respective time period, we found that 16 countries were above the average pre 2000 and 6 post 2000, with Lithuania, Hungary, Serbia, Slovenia and Croatia appearing in both time periods. Additionally, looking at the country's location, we can see that majority of them are from Europe, however, this may be since we have more information in the dataset related to European countries. More concerningly, not surprisingly as 75+ age group is most affected age group overall, is that for both time periods, 75+ age group is the most affected. 13/16 countries pre 2000 had 75+ age group above the mean, whilst post 2000, 5/6 countries had 75+ age group above the mean.

Our dataset had values up-to 2016. Using forecasting, the overall suicide rate per 100k Population decreases, as expected from the overall trend. Looking at, figure 12, we see that not all the age groups are in line with the overall expected trend, with 55–74-year-old age group expected to increase whilst 35–54 age group is expected to see the largest decrease. As the root mean squared error (RSME) is low for the all the age groups, ranging from 0.07-1.31, this model can be reliable in order to predict the future suicide rate.

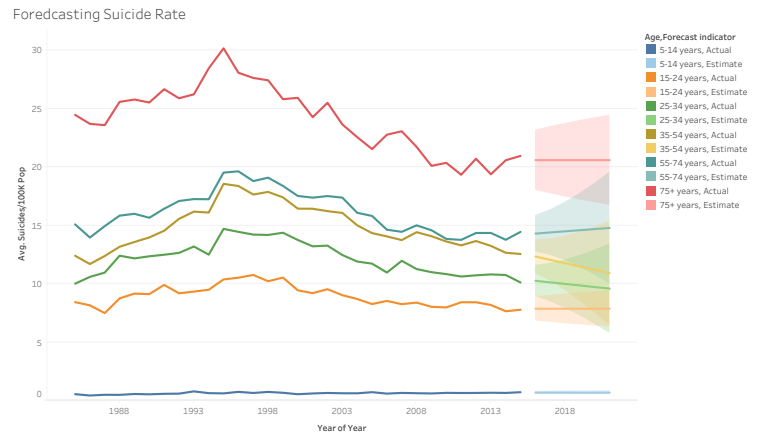


Figure 12 – Shows the forecasted suicide rate for each age group.

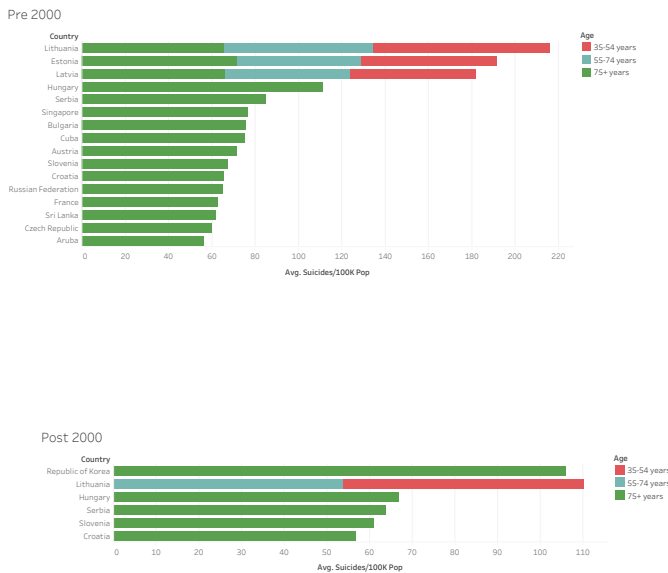


Figure 11- (Left) – Shows countries which are above the above the average, in their respective time period, with age being taken into consideration as well.

4.3 Results

Using time series, we found that there was an overall general increase in suicide rate between the years 1985-1995, then followed by a decrease from 1995 onwards. The country which has been most affected and still is being affected by suicides in terms of population is Lithuania.

Men have been more affected by suicide compared with women. In terms of age, we found that the older the person gets the more suicide rate increased by showing a linearly relationship.

The overall mean suicide rate decreased as you entered post 2000, as well as that that, found that the 75+ age group was the most present in both time periods.

How well a country performs and where it is on the economic cycle has little relationship with suicide rate, with each country having its own unique relationship with changes in GDP that cannot always be explained.

We predict suicide rate to see a overall decrease in suicide rate, with 55–74-year-old to see an increase.

5 CRITICAL REFLECTION

The time series analysis and use of variety of charts used in this paper worked well in identifying trends and patterns, which were made clear with the colour scheme and choice of scales. Having had more information with regards to countries, especially from countries in Asia and Africa, geospatial visualisation would have been an effective technique to visualise changes in the suicide data over time, this would enable to analyse patterns via continents as well as finding anomalies.

Moreover, as the data incorporates values form a large range of years, 30 years, we cannot be sure the method for collecting the information has remained the same, hence the values which we have used possible be inaccurate. In addition to this, when comparing GDP with suicide rate, it showed quite an erratic output which may be because the dataset may not have been real GDP, hence the GDP value we used was not adjusted for inflation. Using GDP values adjusted for inflation may give clearer results.

We used a dataset which gave an overview regarding particular state of a economy and suicide rate but did not contain information specifically related to the individual for example ethnicity, sexual orientation. These factors may able to provide significant patterns with suicide rate hence could potentially provide government information on who is the most venerable hence requite help. Comparing these factors with changes with suicide rate, during changes in economic activity will also provide government information if it needs to provide further help to the particular individual

Table of word counts

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Analysis: Process	1221
Analysis: Results	163
Critical reflection	261

REFERENCES

The list below provides examples of formatting references.

- [1] Xiao, Yunyu, Alvin Junus, Tianzi Li, and Paul Yip. 'Temporal and Spatial Trends in Suicide-Related Visits before and during the COVID-19 Pandemic in the US, 2018–2021'. *Journal of Affective Disorders* 324 (1 March 2023): 24–35. <https://doi.org/10.1016/j.jad.2022.12.062>
- [2] Blasco-Fontecilla, Hilario, M Mercedes Perez-Rodriguez, Rebeca Garcia-Nieto, Pablo Fernandez-Navarro, Hanga Galfalvy, Jose de Leon, and Enrique Baca-Garcia. 'Worldwide Impact of Economic Cycles on Suicide Trends over 3 Decades: Differences According to Level of Development. A Mixed Effect Model Study'. *BMJ Open* 2, no. 3 (1 January 2012): e000785. <https://doi.org/10.1136/bmjopen-2011-000785>.
- [3] Suicide worldwide in 2019: global health estimates. Geneva: World Health Organization; 2021. Licence: CC BY-NC-SA 3.0 IGO. <https://apps.who.int/iris/bitstream/handle/10665/341728/9789240026643-eng.pdf?sequence=1>
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- [5] <https://www.investopedia.com/terms/g/gdp.asp>
- [6] <https://www.investopedia.com/articles/economics/08/past-recessions.asp> (recession dates)
- [7] Hannah Ritchie, Max Roser and Esteban Ortiz-Ospina (2015) - "Suicide". Published online at OurWorldInData.org. Retrieved from: 'https://ourworldindata.org/suicide' [Online Resource]

